

Example Worksheet: The Kitchen Sink

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Video: <https://www.youtube.com/watch?v=dQw4w9WgXcQ>

1 Prerequisites

- Comfort with probability on finite sets ($\mathbb{E}$, distributions on a finite $\mathcal{X}$).
- One variable of calculus: we differentiate $-t \log t$ once.
- From [Prerequisites \(day 0\)](#): big- O notation and the idea of a bit.

What you'll learn

What

- Define the entropy of a distribution and compute it by hand for small examples.
- State and prove Gibbs' inequality, and use it to bound entropy.

- Recognise every markup element the $\text{\LaTeX}$ $\rightarrow$ web pipeline supports, and copy this file as the starting point for a sheet.

Why

Entropy is the unit of account for everything later in the course that measures information: description length, KL divergence, mutual information. A participant who cannot manipulate it fluently is stuck from day two onwards.

How

A morning lecture states the definitions and proves Gibbs' inequality; an exercise session makes the participants reprove it and find the equality case themselves; a reading and discussion block connects it to coding.

2 Roadmap for today

- 10:00 – 11:00 [Lecture: entropy](#)
- 11:00 – 11:15 Break
- 11:15 – 12:30 [Exercise session](#)
- 12:30 – 13:30 Lunch
- 13:30 – 15:00 [Reading and discussion](#)
- 15:00 – 15:30 [Daily checkpoint](#) and feedback form

Required materials

Projector and whiteboard. Print the exercise sheet without solutions (the `-nosol` PDF the site offers) for anyone who wants paper.

3 Content

3.1 Fast track

To get the core in an hour: read Definition [3.1](#) and Theorem [3.2](#) and the proof, do Exercise [3.1](#), then skim [[1](#), ch. 4].

3.2 Main content

The day runs as three sub-modules — a lecture, an exercise session, and a reading block — closing with the daily checkpoint.

3.2.1 Lecture: entropy

Intent

Give the participants the definition and one inequality they can prove themselves, so the afternoon exercises are a rehearsal and not a first sighting. Everything after the proof of Theorem 3.2 can be cut if time is short.

Teaching note

TN, slide 12: draw $-t \log t$ on the board before showing the plot; most participants have never seen that it is concave.

Ordinary prose converts verbatim, including *emphasis*, **bold**, `monospace`, and special characters like 50% or §3. Inline math is copied byte-for-byte: $e^{i\pi} + 1 = 0$, and KaTeX handles comparisons like $x_{<t}$ without trouble. Custom notation defined in the preamble just works: $D_{\text{KL}}(p \parallel q) \geq 0$, $\arg \max_a \mathbb{E}[R \mid a]$, for p, q distributions on $\mathbb{R}^n$. A literal dollar in prose is \$1,000.

Display math works as

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

and as numbered equations you can reference:

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1}$$

Multi-line derivations use `align`:

$$(a+b)^2 = (a+b)(a+b) \tag{2}$$

$$= a^2 + 2ab + b^2. \tag{3}$$

Cross-references resolve to the exact text LaTeX prints: Equation (1) (or, number only, (1)) sits in Section 3.2.1; below we meet Definition 3.1 and Theorem 3.2. Always `\cref`, never a hand-written “Section 3” next to a `\ref` — the build flags it.

Definition 3.1 (entropy). The *entropy* of a distribution p on a finite set $\mathcal{X}$ is

$$H(p) = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

Theorem 3.2 (Gibbs’ inequality). For any distributions p, q on $\mathcal{X}$, $-\sum_x p(x) \log q(x) \geq H(p)$, with equality iff $p = q$.

Proof. By definition, $D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$ by Jensen’s inequality applied to the concave function $\log$; rearranging gives the claim. Equality in Jensen requires

$p(x)/q(x)$ constant, i.e. $p = q$. (On the website this proof is collapsible, like a solution.) □

Lemma 3.3. $H(p) \leq \log |\mathcal{X}|$, with equality iff p is uniform.

Proof. Apply Theorem 3.2 with q uniform. □

Example 3.4. A fair coin has $H = \log 2$: one bit. A coin with $p(\text{heads}) = 1$ has $H = 0$.

Warning

Callouts flag pitfalls, tips, or side notes. Types: **note** (default), **tip**, **warning**. They may be labelled and referenced, like this one. Mind the base of the logarithm: bits for $\log_2$, nats for $\ln$.

A titled callout

A second optional argument titles the box: it replaces the “Tip” label on the PDF frame and heads the box on the web. Handy for a highlighted key equation, $H(p) = -\sum_x p(x) \log p(x)$, or a named aside. Keep maths out of the title itself.

Remark (Continuous distributions)

A **remark** is an aside; on the website it renders as a note-style callout. The optional title is appended in parentheses. Referencing works: see Callout 3.1 above (and this remark is Remark 3.1).

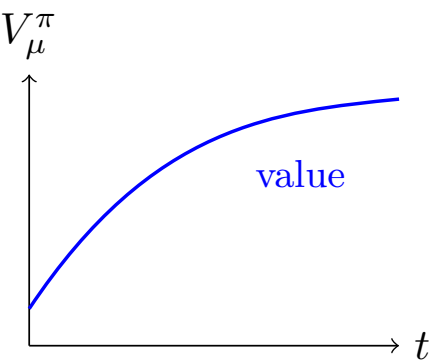
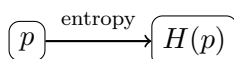


Figure 1: An example figure from `fig/`. On the web this becomes a **Figure** component.

See Figure 1. The same in TikZ, inline:



Citations come from `biblo.bib`: the idea of entropy is due to [2]; for a textbook treatment see [1, ch. 4]. On the web every citation becomes author–year text linking to a References list at the foot of the page, whatever `\bibliographystyle` the PDF uses. External links are plain [hyperlinks](#), and footnotes are supported.¹

3.2.2 Exercise session

Intent

The participants reprove the lecture’s inequality with their own hands and find its equality case, so that “entropy is maximised by the uniform distribution” becomes something they derived rather than heard. Pairs; Exercise 3.2 is the one to discuss on the board.

Exercise 3.1 (Entropy warm-up). Let p be a distribution on a finite set $\mathcal{X}$.

- (a) Show that $H(p) \geq 0$.
- (b) For which p is $H(p) = 0$?

Solution to Exercise 3.1

For Exercise 3.1(a): each term $-p(x) \log p(x) \geq 0$ since $p(x) \in [0, 1]$. For part (b): $H(p) = 0$ iff every term vanishes iff p is a point mass. [*Hint*: when is $-t \log t = 0$ on $[0, 1]$?]

Exercise 3.2 (Gibbs, the hard direction). (★) Prove the equality case of Theorem 3.2 directly, without citing Jensen’s equality condition. [*Note*: this one is solved at the end of the sheet instead of inline, to demonstrate both solution placements.]

Hint: Consider $f(t) = t - 1 - \log t$: it is nonnegative on $t > 0$ and vanishes only at $t = 1$.

Exercise 3.3 (Open-ended). Why might a learning system that compresses its inputs well also predict them well? No solution is given: this is a discussion exercise.

¹On the web this becomes a numbered footnote: the marker links down to a note at the foot of the page, and the note links back. In the PDF it is an ordinary L^AT_EX footnote.

3.2.3 Reading and discussion

Intent

Connect the morning's definition to coding, through one well-curated reading with clear take-aways, then a discussion that compares and contrasts. Groups of three to four; steer towards the source-coding theorem if the discussion stalls.

- [1, ch. 4], sections 4.1–4.3 (about 40 minutes). Take-away: the entropy is the average length of the best code.
- [2], section 6 only. Take-away: where the axioms for H come from.

Discussion prompt

Two participants disagree: one says entropy is a property of the source, the other that it is a property of the observer's beliefs about the source. Who is right, and does Theorem 3.2 settle it?

3.2.4 Daily checkpoint

A ten-minute quiz, [here](#), then the daily feedback form.

Notes for future iterations

The reading block ran long last time; consider dropping Shannon's section 6 and giving the discussion prompt the extra half hour.

3.3 Learn more

Deeper. [1, ch. 5–6] for source coding and the noisy channel. **Frontier.** The [minimum description length](#) literature, which is where this thread is picked up on the Solomonoff induction day.

A Solutions

Solution to Exercise 3.2

Write $f(t) = t - 1 - \log t$ for $t > 0$ and note $f(t) \geq 0$ with equality iff $t = 1$. Then with $t_x = q(x)/p(x)$,

$$-\sum_x p(x) \log q(x) - H(p) = \sum_x p(x) f(t_x) + \sum_x (p(x) - q(x)) \geq 0,$$

and equality forces $t_x = 1$ wherever $p(x) > 0$, i.e. $p = q$.

Instructor note. Exercise 3.2 is the one to discuss on the board if time is short; the warm-up can be left for self-study.

References

- [1] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [2] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.